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The Editor-in-Chief
IEEE Transactions on Biometrics, Behavior, and Identity Science

Dear Editors,

We submit the manuscript "Part supervision versus part matching for pose-based gait recognition under missing joints" for consideration as a regular paper in *IEEE Transactions on Biometrics, Behavior, and Identity Science*.

Part-aware gait recognizers usually add two things at once: identity losses on body regions during training and fusion of regional similarities at retrieval. Gains are then credited to one of them without testing the other. Our study separates the two. On one fixed pose-graph encoder, seven training objectives, including global and part supervised contrastive losses and cross-entropy controls, were trained with five seeds each on CASIA-B development identities. No auxiliary objective differed reliably from a metric-learning baseline, and an apparent penalty of part supervision in an earlier three-seed experiment did not replicate. In contrast, fusing normalized raw part pools from frozen checkpoints improved corrupted-probe accuracy under every objective. On Gait3D, with the recipe frozen on development identities and the test identities evaluated once, the same matcher raised Rank-1 from 21.88% to 23.86% and reached parity with GPGait, which we reproduced with its official configuration on identical data.

We then applied the matcher to GPGait itself, and the result reversed: GPGait's part-specific branches, trained with their own losses, add 9.1 points on CASIA-B and 5.2 on Gait3D, while raw part matching lowers its accuracy. The paper therefore concludes that where a part-aware gain comes from depends on the encoder, and that global, raw regional and trained regional matching should be compared on the same frozen checkpoints. We report the limitations openly, including a large accuracy gap to GPGait on CASIA-B, earlier project exposure to the CASIA-B test identities, an available-pose subset of Gait3D and the absence of labelled real occlusion.

We believe the paper suits *T-BIOM* because it addresses how part-aware gait biometrics should be evaluated under missing body evidence, and supports its answer with controlled experiments, seed-by-identity bootstrap intervals, reproduced baselines on identical data and a transfer test on an established encoder. Every table and figure can be regenerated from the saved per-probe outcomes with the released scripts.

This manuscript is original, has not been published, is not under consideration for publication elsewhere, and does not extend a conference paper. It uses only publicly released datasets, and no new data involving human participants were collected (see the Ethics Statement). Both authors approved the submission and declare no competing interests. The study received no specific funding. The use of an AI assistant during manuscript preparation is disclosed in the Acknowledgment,

following IEEE policy.

Sincerely,

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